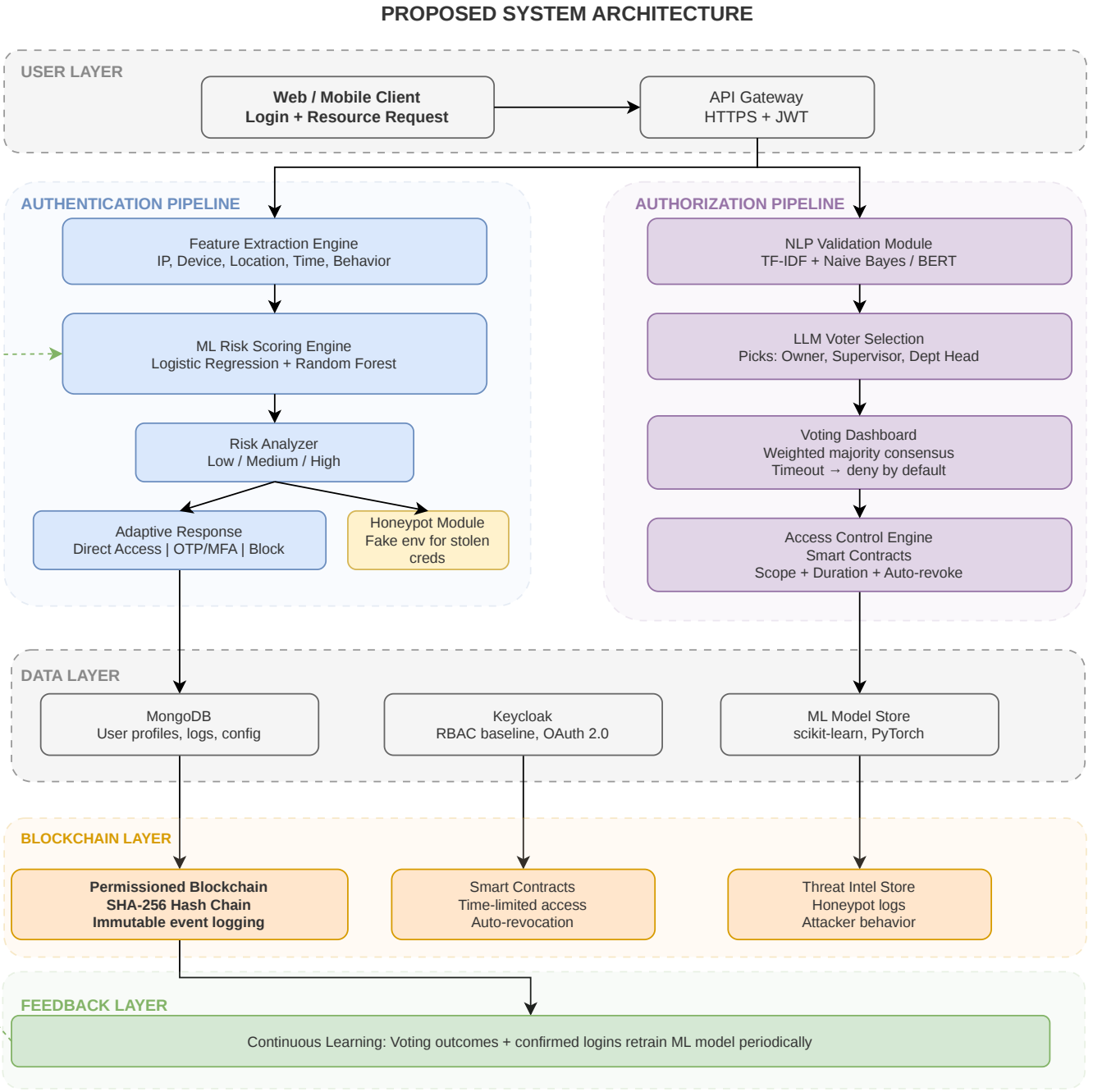

PROPOSED SYSTEM ARCHITECTURE
USER LAYER
Web / Mobile Client
Login + Resource Request
API Gateway
HTTPS + JWT
AUTHENTICATION PIPELINE
Feature Extraction Engine
IP, Device, Location, Time, Behavior
ML Risk Scoring Engine
Logistic Regression + Random Forest
Risk Analyzer
Low / Medium / High
Adaptive Response
Direct Access | OTP/MFA | Block
Honeypot Module
Fake env for stolen creds
AUTHORIZATION PIPELINE
NLP Validation Module
TF-IDF + Naive Bayes / BERT
LLM Voter Selection
Picks: Owner, Supervisor, Dept Head
Voting Dashboard
Weighted majority consensus
Timeout → deny by default
Access Control Engine
Smart Contracts
Scope + Duration + Auto-revoke
DATA LAYER
MongoDB
User profiles, logs, config
Keycloak
RBAC baseline, OAuth 2.0
ML Model Store
scikit-learn, PyTorch
BLOCKCHAIN LAYER
Permissioned Blockchain
SHA-256 Hash Chain
Immutable event logging
Smart Contracts
Time-limited access
Auto-revocation
Threat Intel Store
Honeypot logs
Attacker behavior
FEEDBACK LAYER
Continuous Learning: Voting outcomes + confirmed logins retrain ML model periodically